Service Bulletin

FILING INSTRUCTIONS	
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MODELS AFFECTED: 327B, 330B, 335B, and 340B Articulated Dump Trucks

SUBJECT: Loss of Nitrogen Charge Pressure in the Front Suspension Cylinders

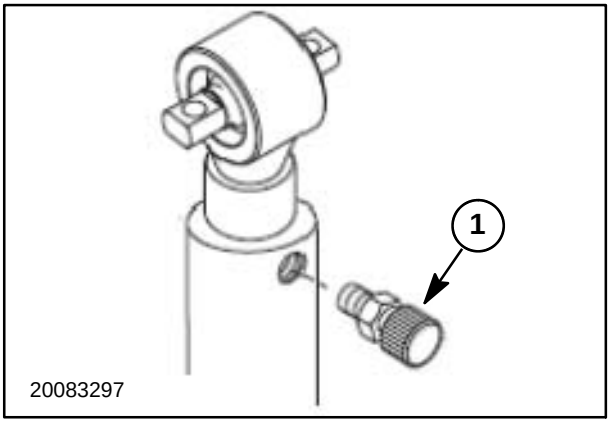
DESCRIPTION OF CONCERN

The front suspension cylinders act as shock absorbers to allow the machine to ride smoothly over rough terrain. The cylinders are filled with hydraulic fluid and have a nitrogen charge to prevent impact to the frame.

If the nitrogen charge is lost due to leakage, the cylinders do not operate correctly. This can result in an uneven ride and / or stress on the frame and the attached components.

ACTION/CORRECTION

A new inflation valve (part number 8998341362), 1, is available that incorporates a stronger spring to prolong the nitrogen charge in the cylinders. The new inflation valve reduces the nitrogen loss that can occur while the machine is operated over rough terrain or in other severe applications.

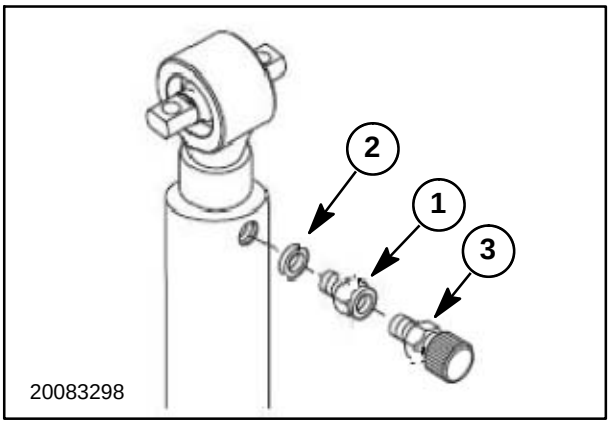


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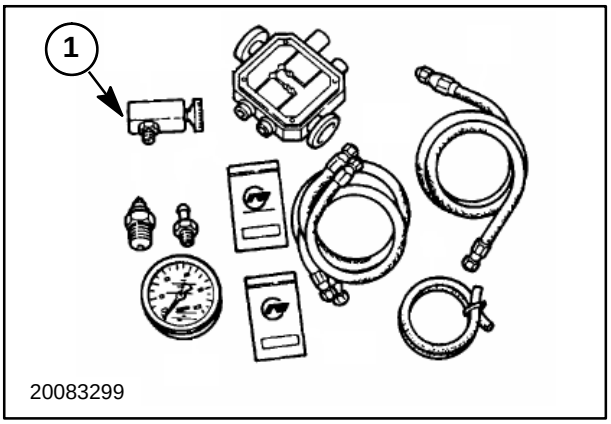
These changes were also made to support the new inflation valve:

- An adapter (part number 89500431250), 1, and an O-ring (part number 8998341923), 2, are needed to install the new inflation valve, 3, onto the existing cylinders (part number 89500415397).
- Improved cylinders (part number 89500431196) are available that use the new inflation valve. The improved cylinders do not require the adapter and the O-ring with the new inflation valve.



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- A new check device (part number 8900046374), 1, that opens the new inflation valve is included in the updated suspension inflation kit (part number 8900127814). The check device is also available separately for use with earlier suspension inflation kits.



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It is recommended that a copy of this Service Bulletin be put into the appropriate Parts Catalogs/Repair Manuals for reference.

Production:

The new inflation valve (part number 8998341362) and the improved cylinders (part number 89500431196) were introduced into production at these serial numbers:

Model	Serial Number
327B	N8PG56202
330B	
335B	N8PG58085
340B	

Service:

Update the existing cylinders (part number 89500415397) with the new inflation valve, adapter, and O-ring.

NOTE: *Perform this procedure for both cylinders at the same time.*

1. Secure the machine and obtain access to the cylinders. For detailed instructions, refer to the suspensions section in the repair manual for your model.
   - Ensure that the machine has been parked for at least one hour.
   - Ensure that the front of the machine is raised off of the ground to extend the suspension to the end of the mechanical stop.
2. Release the pressure from the cylinder. For detailed instructions, refer to the suspensions section in the repair manual for your model.
3. Remove the existing inflation valve (part number 8900132337) from the cylinder.
4. Apply Loctite® 557 to the threads of the adapter (part number 89500431250).
5. Install the O-ring (part number 8998341923) and the adapter where you removed the existing inflation valve.
6. Install the new inflation valve into the adapter.
7. Correctly inflate the cylinders. For detailed instructions, see ‘Inflate Cylinders’ attached at the end of this bulletin.

NOTE*: Once the new inflation valve is installed, the service information in the repair manual is obsolete. Updated service information for cylinders with the new inflation valve is provided at the end of this bulletin.*

8. Return the machine to operational status.

ADDITIONAL INFORMATION

Updates to the suspension inflation kit and to the service procedures are available to support the new inflation valve. See ‘Updated Service Information’ attached at the end of this bulletin.

PARTS AFFECTED

Previous Part #	Qty	Description	Qty	New Part #	SupportPRO2 Ref Number
89500415397	2	Assembly, cylinder, front suspension	2	89500431196	38D00000017 (001)
8900132337	2	Valve, inflation	2	8998341362	38D00000017 (001)
NA	NA	Adapter[1]	2	89500431250	NA
NA	NA	O-Ring[1]	2	8998341923	NA
89500409614	1	Kit, suspension inflation	1	8900127814	6.769A (03)
NA	NA	Device, check[2]	1	8900046374	6.769A (03)

[1]This part is only required to update the existing cylinders (part number 89500415397) with the new inflation valve (part number 8998341362).

[2]This part is included in the updated suspension inflation kit (part number 8900127814). This part is also available separately to use with an earlier suspension inflation kit.

WARRANTY STATUS

This service bulletin is for informational purposes only and does not constitute warranty authorization to update units without a warrantable failure.

UPDATED SERVICE INFORMATION

This document applies to any articulated dump truck that has the new inflation valve installed on the front suspension cylinders.

See the information that follows for:

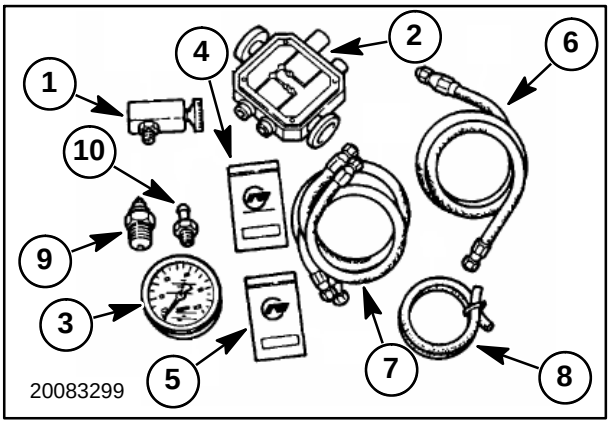
- Updated suspension inflation kit.
- Updated service procedures.
- Service interval and specifications.

UPDATED SUSPENSION INFLATION KIT

See figure 4 and the table below for the components of the updated suspension inflation kit.

Item Number	Description	Qty.
1	Check device	1
2	Inflation device	1
3	Pressure gauge, 0 to 50 bar (0 to 725 psi)	1
4	Washer, copper	4
5	Washer, copper	4
6	Hose	1
7	Hose	1
8	Tube, clear	1
9	Fitting	1
10	Fitting	1

NOTE: *Earlier suspension inflation kits did not include the check device. For earlier kits, the check device (part number 8900046374) is available separately through service parts.*



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UPDATED SERVICE PROCEDURES

NOTE: *Perform service procedures for both cylinders at the same time.*

These updated procedures support the new inflation valve:

- Check the cylinder height.
- Check the nitrogen charge pressure.
- Inflate cylinders.
- Deflate cylinders.
- Change the hydraulic fluid.

NOTE: *The updated procedures supersede the procedures in the repair manual.*

Secure the Machine

Follow these steps before you service the cylinders:

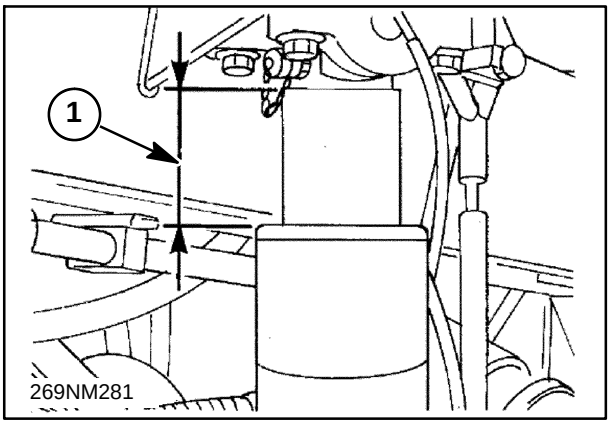
1. Empty all material from the dump body.
2. Park the machine on a level surface so that the front and the back of the machine are in a straight line.
3. Use the steering lock bar to block the articulation.
4. Turn the ignition key to the 'OFF' position to shut down the engine.
5. Leave the machine parked for one hour before performing service.

Check the Cylinder Height

1. If you have not already done so, secure the machine. For detailed instructions, see 'Secure the Machine' earlier in this document.

NOTE: *Ensure that the machine has been parked at least one hour before proceeding.*

2. Measure the height, 1, of the cylinder. See the last page of this document for the cylinder height specifications.
3. If the measured height meets specifications, restore the machine to operational status.
4. If the measured height does not meet specifications, check the nitrogen charge pressure for the front suspension cylinders. For detailed instructions, see 'Check the Nitrogen Charge Pressure' later in this document.

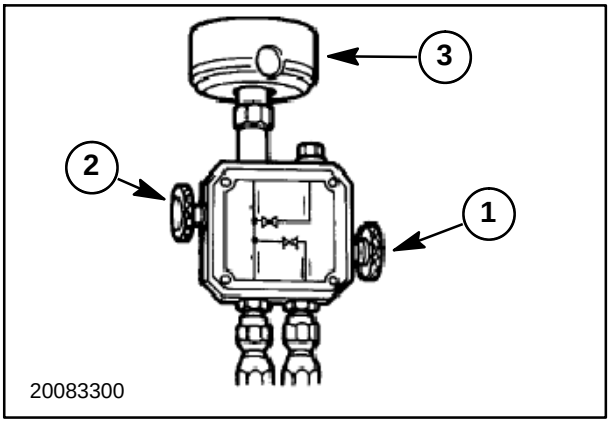


Check the Nitrogen Charge Pressure

1. If you have not already done so, secure the machine. For detailed instructions, see ‘Secure the Machine’ earlier in this document.

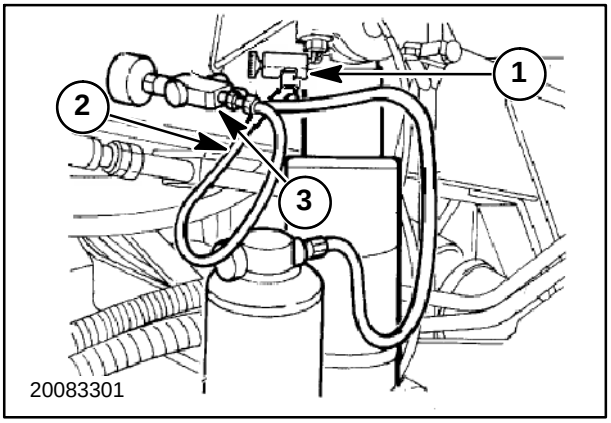
NOTE: *Ensure that the machine has been parked at least one hour before proceeding.*

2. Remove the lower bolt and the end-stroke rope to allow the maximum cylinder stroke.
3. Use a suitable lifting device (for example, a hydraulic jack) to raise the front tires off of the ground and fully extend the cylinders.
4. Ensure that the IN knob, 1, and the OUT knob, 2, on the inflation device are closed.
5. Attach the pressure gauge, 3, onto the inflation device.



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6. Remove the cap from the inflation valve.
7. Insert the fitting and the copper washers, if needed, into the inflation valve.
8. Insert the check device, 1, onto the fitting.
9. Attach a hose, 2, to the check device and to the inflation device, 3, with the pressure gauge.



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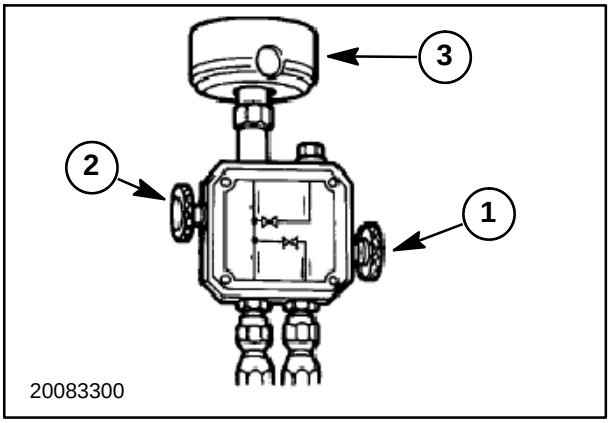
10. Slowly turn the check device knob clockwise to open the inflation valve.
11. Identify the pressure indicated on the pressure gauge.
12. If the pressure is within specifications:
    - **a.** Slowly turn the check device knob counter-clockwise to close the inflation valve.
    - **b.** Remove the check device and the fitting.
    - **c.** Reinstall the cap onto the inflation valve.
    - **d.** Restore the machine to operational status.
13. If the pressure is not within specifications, add or purge nitrogen as needed. For detailed instructions, see ‘Inflate Cylinders’ or ‘Deflate Cylinders’, later in this document.

Inflate Cylinders

1. If you have not already done so, secure the machine. For detailed instructions, see 'Secure the Machine' earlier in this document.

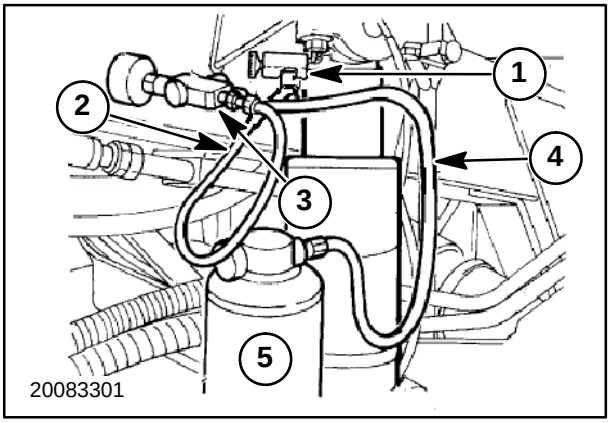
NOTE: *Ensure that the machine has been parked at least one hour before proceeding.*

2. Remove the lower bolt and the end-stroke rope to allow the maximum cylinder stroke.
3. Use a suitable lifting device (for example, a hydraulic jack) to raise the front tires off of the ground and fully extend the cylinders.
4. Ensure that the IN knob, 1, and the OUT knob, 2, on the inflation device are closed.
5. Attach the pressure gauge, 3, onto the inflation device.



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6. Remove the cap from the inflation valve.
7. Insert the fitting and the copper washers, if needed, into the inflation valve.
8. Insert the check device, 1, onto the fitting.
9. Attach a hose, 2, to the check device and to the inflation device, 3, with the pressure gauge.
10. Attach the other hose, 4, to the nitrogen supply bottle, 5, and to the other hose port on the inflation device.



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11. Slowly turn the check device knob clockwise to open the inflation valve.
12. Open the nitrogen supply bottle.
13. Slowly turn the IN knob on the inflation device counter-clockwise. Nitrogen flows into the cylinder.
14. When the pressure reaches 38 bar (551 psi) turn the IN knob clockwise to close.
15. Close the nitrogen supply bottle.

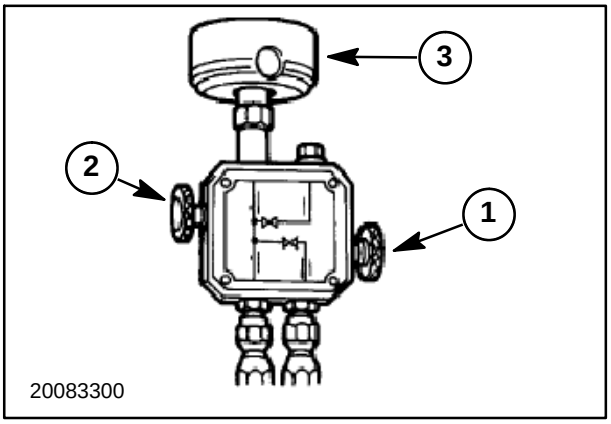
16. Check the pressure reading.
17. If the pressure is above specifications:
    a. Turn the OUT knob on the inflation device counter-clockwise to lower the pressure.
    b. When the desired pressure is met, turn the OUT knob clockwise to close.
18. Slowly turn the check device knob counter-clockwise to close the inflation valve.
19. Remove the check device and the fitting.
20. Reinstall the cap onto the inflation valve.
21. Restore the machine to operational status.

Deflate Cylinders

1. If you have not already done so, secure the machine. For detailed instructions, see 'Secure the Machine' earlier in this document.

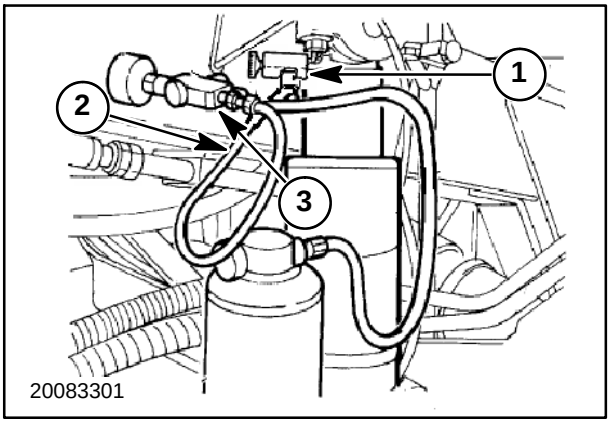
NOTE: *Ensure that the machine has been parked at least one hour before proceeding.*

2. Remove the lower bolt and the end-stroke rope to allow the maximum cylinder stroke.
3. Use a suitable lifting device (for example, a hydraulic jack) to raise the front tires off of the ground and fully extend the cylinders.
4. Ensure that the IN knob, 1, and the OUT knob, 2, on the inflation device are closed.
5. Attach the pressure gauge, 3, onto the inflation device.



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6. Remove the cap from the inflation valve.
7. Insert the fitting and the copper washers, if needed, into the inflation valve.
8. Insert the check device, 1, onto the fitting.
9. Attach a hose, 2, to the check device and to the inflation device, 3, with the pressure gauge.



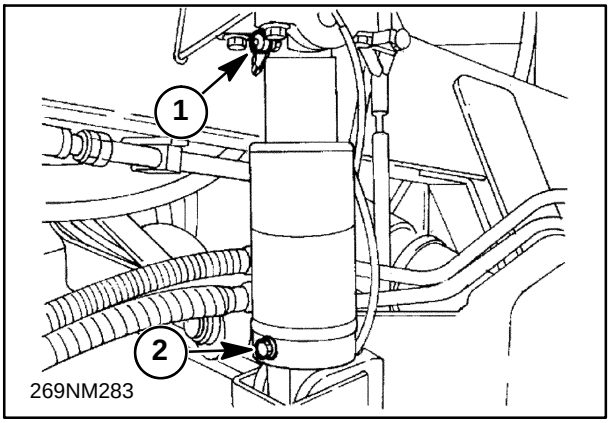
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10. Slowly turn the check device knob clockwise to open the inflation valve.
11. Slowly turn the OUT knob on the inflation device counter-clockwise to release the nitrogen charge pressure.
12. Watch the pressure gauge reading. When the charge pressure meets specifications, close the OUT knob.
13. Slowly turn the check device knob counter-clockwise to close the inflation valve.
14. Remove the check device and the fitting.
15. Reinstall the cap onto the inflation valve.
16. Restore the machine to operational status.

Check the Hydraulic Fluid

NOTE: *Ensure that you collect and dispose of the hydraulic fluid in accordance with local laws and / or regulations.*

1. If you have not already done so, secure the machine. For detailed instructions, see 'Secure the Machine' earlier in this document.
2. Remove the lower bolt and the end stroke rope to allow the maximum cylinder stroke.
3. Use a suitable lifting device (for example, a hydraulic jack) to raise the front tires off of the ground and fully extend the cylinders.
4. Remove the nitrogen charge pressure. For detailed instructions, see 'Deflate Cylinders' earlier in this document.
5. Remove the inflation valve, 1.
6. Remove the drain cap, 2, and allow the hydraulic fluid to drain into the collection container.



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7. Clean and inspect the drain cap. Replace as needed.
8. When the hydraulic fluid has completely drained, replace the O-ring and reinstall the drain cap.
9. Add the hydraulic fluid at the hole for the inflation valve.
10. Fill the cylinder within specifications.
11. Reinstall the inflation valve.
12. Restore the nitrogen charge pressure. For detailed instructions, see 'Inflate Cylinders' earlier in this document.
13. Restore the machine to operational status.
14. Operate the machine and travel a short distance.
15. Ensure that the cylinder height meets specifications.

SERVICE INTERVAL AND SPECIFICATIONS

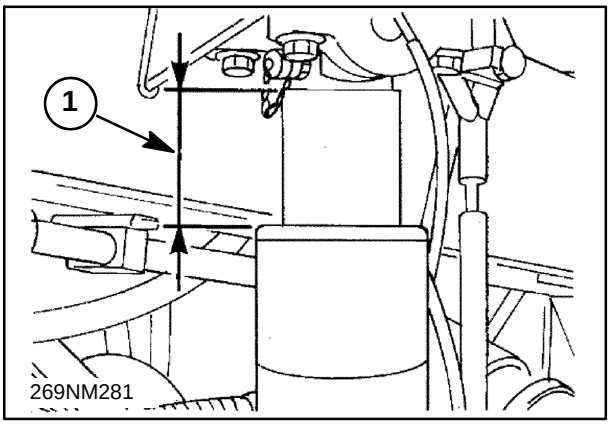
Service Interval

Operation	Daily	100 Hrs	500 Hrs	2000 Hrs
Visually check for leakages	X			
Check element height		X	X	X
Check element pressure			X	X
Oil/gas replacement				X

Specifications

Models 327B and 330B

Stem diameter	100 mm (3.94 inch)
External diameter	150 mm (5.91 inch)
Internal diameter (bore)	125 mm (4.92 inch)
Center-to-center with cylinder closed	506 mm (19.92 inch)
Maximum stroke	120 mm (4.72 inch)
Internal volume with cylinder closed	1705 cm^3 (104.05 $inch^3$)
Maximum working pressure	200 bar (2901 psi)
Inflation pressure (fully extended)	30 bar (435 psi)
Fluid quantity	1.55 L (0.41 US gal)
Check height, 1*	57 to 87 mm (2.24 to 3.43 inch) **Recommended:** 72 mm (2.83 inch)
*Check the height when the machine is empty and has been parked for at least one hour. Measure the height between the start of the chromed stem to the cylinder end as shown in figure 13.	



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Models 335B and 340B

Stem diameter	100 mm (3.94 inch)
External diameter	150 mm (5.91 inch)
Internal diameter (bore)	125 mm (4.92 inch)
Center-to-center with cylinder closed	506 mm (19.92 inch)
Maximum stroke	120 mm (4.72 inch)
Internal volume with cylinder closed	1705 cm^3 (104.05 $inch^3$)
Maximum working pressure	200 bar (2901 psi)
Inflation pressure (fully extended)	37 bar (537 psi)
Fluid quantity	1.55 L (0.41 US gal)
Check height, 1*	75 to 105 mm (2.95 to 4.13 inch) **Recommended:** 90 mm (3.54 inch)
*Check the height when the machine is empty and has been parked for at least one hour. Measure the height between the start of the chromed stem to the cylinder end as shown in figure 13.	